

**2.10**

Inverses of Exponential Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Exponential domain and range swap to become logarithmic range and domain.

2. Write one precise sentence explaining this principle: The horizontal exponential asymptote becomes a vertical logarithmic asymptote after reflection.

3. Write one precise sentence explaining this principle: To find an inverse, isolate the exponential expression before taking a logarithm.

4. Write one precise sentence explaining this principle: Composition verifies the inverse on appropriate domains.

Core

5. Find the inverse of $f(x) = 2^{x-3} + 5$.
6. Identify the vertical asymptote of the inverse above.
7. Verify $\log_3(3^x) = x$.
8. State what the formula $y = ab^{x-h} + k \Rightarrow x = h + \log_b\left(\frac{y-k}{a}\right)$ tells you and name any required restriction.
9. State what the formula $f^{-1}(x) = h + \log_b\left(\frac{x-k}{a}\right)$ tells you and name any required restriction.
10. Create a second representation (table, graph, formula, or context) for one central relationship in Inverses of Exponential Functions. State its domain and justify one key feature.

Medium

11. Find the inverse of $f(x) = 2^{x-1} + 3$, including its domain.
12. Find the inverse of $f(x) = 5 \cdot 3^x - 2$, including its domain.
13. Find the inverse of $f(x) = -4(1/2)^x + 7$, including its domain.

14. Find the inverse of $f(x) = e^{2x+1}$, including its domain.

15. Find the inverse of $f(x) = 10^{x/3}$, including its domain.

Hard

16. Original exponential has horizontal asymptote $y=5$. What asymptote does its inverse have?

17. Find inverse of $f(x) = 3^{2-x}$ and determine whether inverse is increasing.

18. Verify inverse pair $2^x + 1$ and $\log_2(x - 1)$.

19. Explain why the inverse of an increasing exponential is increasing.

Difficult

20. Find parameters a, k if $f(x) = a2^x + k$ and $f^{-1}(x) = \log_2(x - 3) - 1$.

21. Find inverse of $f(x) = \frac{e^x}{1+e^x}$, $0 < f < 1$.

22. Prove horizontal translations of an exponential become vertical translations of its inverse in the opposite coordinate role.

Challenge / HOT

23. Construct an exponential whose inverse has vertical asymptote $x = -4$ and passes through $(5, 2)$.

24. Can an exponential with negative multiplier have a logarithmic inverse? Explain domain/range.

AP-Style Multiple Choice

25. Inverse of 2^x is

- (A) $1/2^x$
 - (B) $\log_2 x$
 - (C) x^2
 - (D) e^x
26. Horizontal asymptote $y=k$ becomes inverse
- (A) horizontal $y=k$
 - (B) vertical $x=k$
 - (C) x-axis
 - (D) none
27. Inverse domain equals exponential
- (A) domain
 - (B) range
 - (C) slope
 - (D) base
28. Inverse of $e^x + 4$ is
- (A) $\ln x + 4$
 - (B) $\ln(x - 4)$
 - (C) e^{x-4}
 - (D) $1/(e^x + 4)$
29. Exponential with $b>1$ has inverse that is
- (A) decreasing
 - (B) increasing
 - (C) constant
 - (D) periodic
30. To invert, first
- (A) square both sides
 - (B) isolate exponential expression
 - (C) set $x=0$
 - (D) differentiate

AP-Style Free Response

- 31.** Let $f(x) = 6 - 2 \cdot 3^{x+1}$. (a) Find inverse. (b) State original range and inverse domain. (c) Find inverse asymptote. (d) Verify a point pair.

32. An inverse model is $t(A) = 5 \ln((A - 20)/80)$. (a) Find original $A(t)$. (b) State domains. (c) Interpret 20 and 80.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.